AI RESPONSE EVALUATION

Independent Portfolio Exercise | 20 Illustrative Paired Cases

Prepared for generalist AI evaluation practice • October 2026

Project overview

This portfolio demonstrates how to apply a consistent rubric to paired candidate responses. All prompts and responses below are AUTHOR-CREATED HYPOTHETICAL EXAMPLES, not outputs collected from any AI system. Results must not be represented as measured AI model performance.

Evaluation rubric

- Accuracy: Is the content factually correct?
- Reasoning: Is the logic valid and supported?
- Instruction adherence: Does the response meet format and content constraints?
- Completeness: Does it address all requested parts?
- Clarity: Is it easy to understand and unambiguous?

Score each criterion from 0 (fails), 1 (partially meets), to 2 (fully meets); maximum 10 points per response. Use N/A only where a criterion genuinely does not apply, and normalize scores when comparing. Preferred response is selected by correctness and task fit, not by verbosity.

Illustrative cases and evaluator decisions

01. Arithmetic

Prompt: What is 18 × 7?

Response A: 126

Response B: 125

Preferred: A | Evaluation note: Response B contains a calculation error.

02. Arithmetic

Prompt: What is 15% of 200?

Response A: 30

Response B: 25

Preferred: A | Evaluation note: Response B contains a calculation error.

03. Unit conversion

Prompt: How many minutes are in 2.5 hours?

Response A: 150 minutes

Response B: 120 minutes

Preferred: A | Evaluation note: Response B uses the wrong duration.

04. Reasoning

Prompt: All roses are flowers. Does that imply all flowers are roses?

Response A: No, the converse does not follow.

Response B: Yes, because roses are flowers.

Preferred: A | Evaluation note: Response B makes a logical or arithmetic error.

05. Reasoning

Prompt: A train departs at 09:10 and travels 80 minutes. Arrival?

Response A: 10:30

Response B: 10:20

Preferred: A | Evaluation note: Response B makes a logical or arithmetic error.

06. Instructions

Prompt: Reply with exactly three words: acknowledge the request.

Response A: Request duly acknowledged

Response B: I acknowledge your request.

Preferred: A | Evaluation note: Response B violates the explicit formatting or count constraint.

07. Instructions

Prompt: List exactly two prime numbers under 10.

Response A: 2, 3

Response B: 2, 3, 5

Preferred: A | Evaluation note: Response B violates the explicit formatting or count constraint.

08. Instructions

Prompt: Give a one-sentence definition of photosynthesis.

Response A: Photosynthesis is the process by which plants use light to make sugars from carbon dioxide and water.

Response B: Photosynthesis is how plants grow. It needs sunlight.

Preferred: A | Evaluation note: Response B violates the explicit formatting or count constraint.

09. Accuracy

Prompt: Is the Pacific Ocean larger than the Atlantic?

Response A: Yes.

Response B: No.

Preferred: A | Evaluation note: Response B gives an incorrect fact.

10. Accuracy

Prompt: What is the chemical symbol for sodium?

Response A: Na

Response B: So

Preferred: A | Evaluation note: Response B gives an incorrect fact.

11. Accuracy

Prompt: How many sides does a hexagon have?

Response A: Six

Response B: Eight

Preferred: A | Evaluation note: Response B gives an incorrect fact.

12. Clarity

Prompt: Define a budget for a beginner.

Response A: A budget is a plan for how to earn, spend and save money.

Response B: A budget is a fiscal allocation construct.

Preferred: A | Evaluation note: Response A uses more accessible wording.

13. Completeness

Prompt: Name two causes of soil erosion.

Response A: Heavy rainfall and deforestation.

Response B: Rainfall.

Preferred: A | Evaluation note: Response B omits requested information.

14. Completeness

Prompt: What two things are needed to calculate speed?

Response A: Distance traveled and time taken.

Response B: Distance traveled.

Preferred: A | Evaluation note: Response B omits requested information.

15. Reasoning

Prompt: If a shirt costs £40 after a 20% discount, what was the original price?

Response A: £50, because £40 is 80% of the original.

Response B: £48, because 20% of £40 is £8.

Preferred: A | Evaluation note: Response B makes a logical or arithmetic error.

16. Instructions

Prompt: Answer only yes or no: Is 9 an odd number?

Response A: Yes

Response B: Yes, nine is odd.

Preferred: A | Evaluation note: Response B violates the explicit formatting or count constraint.

17. Accuracy

Prompt: What is the capital of Kenya?

Response A: Nairobi

Response B: Mombasa

Preferred: A | Evaluation note: Response B gives an incorrect fact.

18. Reasoning

Prompt: If 3 notebooks cost £12, how much do 5 cost at the same unit price?

Response A: £20

Response B: £15

Preferred: A | Evaluation note: Response B makes a logical or arithmetic error.

19. Completeness

Prompt: Give two benefits of recycling.

Response A: It can reduce landfill waste and conserve raw materials.

Response B: It is good for the planet.

Preferred: A | Evaluation note: Response B omits requested information.

20. Instructions

Prompt: Provide a numbered list of exactly two ways to save water.

Response A: 1. Fix leaks. 2. Turn off taps when not in use.

Response B: Fix leaks, turn off taps, and collect rainwater.

Preferred: A | Evaluation note: Response B violates the explicit formatting or count constraint.

Findings and limitations

Across these 20 constructed pairs, Response A was intentionally designed to be preferable in every case. This is a demonstration of rubric application, not a blind study or empirical accuracy rate. The exercise includes seven task categories and highlights that correctness, adherence to constraints, and completeness should be assessed separately. A future real benchmark should use actual model outputs, randomized response order, independent reviewers, and inter-rater agreement checks.

How to extend into a real study

- Collect 20–30 authentic outputs from permitted AI systems and record model/version/date.
- Randomize A/B placement, score both independently on all five dimensions, and log justifications.
- Calculate average scores and agreement between two reviewers; report disagreements and limitations.